Credit Risk Analysis

Transforming Raw Data into Actionable Insights

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Project Overview

This project aims to conduct a comprehensive analysis of a financial institution's loan portfolio to understand loan performance, assess risk, and measure overall profitability. The analysis utilized a dataset containing details of thousands of loans, including loan amounts, repayment status, and borrower demographics.

Dataset Description

The databases were obtained from kaggle and contain loan data for a US bank. They comprise 38,576 rows and 24 columns (attributes), that can be analyzed to understand patterns and relationships. Each attribute provides valuable information that contributes to the overall analysis.

Sample of dataset

grade	home_ownership	issue_date	last_credit_pull_date	last_payment_date	...	sub_grade	term	verification_status	annual_income	dti
C	RENT	2/11/2021	9/13/2021	4/13/2021	...	C4	60 months	Source Verified	30000.0	0.0100
E	RENT	1/1/2021	12/14/2021	1/15/2021	...	E1	36 months	Source Verified	48000.0	0.0535
C	RENT	1/5/2021	12/12/2021	1/9/2021	...	C5	36 months	Not Verified	50000.0	0.2088
B	MORTGAGE	2/25/2021	12/12/2021	3/12/2021	...	B2	60 months	Source Verified	42000.0	0.0540
A	MORTGAGE	1/1/2021	12/14/2021	1/15/2021	...	A1	36 months	Verified	83000.0	0.0231

Description of Columns Name

id	A unique identification number for each loan application
member_id	A unique identification number for the borrower (member)
emp_title	The borrower's job title
emp_length	Employment period (number of years of experience)
annual_income	The borrower's annual income
home_ownership	Housing ownership status (rent, mortgage, or ownership)
address_state	The state where the borrower resides
dti (Debt-To-Income Ratio)	Debt-to-income ratio; a measure of a borrower's ability to make monthly payments
loan_amount	The total loan amount required

term	Loan term (36 months or 60 months)
int_rate	Interest rate
installment	Monthly installment amount
grade & sub_grade	credit rating assigned by the bank for the degree of risk of the loan
purpose	Purpose of the loan
loan_status	loan status (Current, Charged Off or Fully Paid)
verification_status	Is the borrower's income verified or not?(Source Verified, Verified or Not Verified)
total_payment	Total amount paid by the borrower to date
total_acc	Total number of credit lines (accounts) opened for the borrower

issue_date	Date the loan was issued or granted
last_payment_date	Date of last payment received from the borrower
next_payment_date	Date of the next scheduled payment
last_credit_pull_date	The last date the bank withdrew the borrower's credit history for review

Insights

Total Loan Applications: 38576
Total Funded Amount: $435.76 M
Total Amount Received: $473.07 M
Avg Int Rate:12.05%
Avg DTI Rate:13.33%
Net profit: $37.31 M
profit margin: 8.56%

Good Loan: its "Fully Paid" and "Current" loans
Good Loan Applications: 33243
Good Loan Funded Amount (in Millions): $370.22M
Good Loan Total Received (in Millions): $435.79M
Percentage of Good Loan Applications: 86.18%

Bad Loan: Charged Off loans
Bad Loan Applications: 5333
Bad Loan Funded Amount (in Millions): $65.53M
Bad Loan Total Received (in Millions): $37.28M
Percentage of Bad Loan Applications: 13.82%

loans fully paid: 83.3%
loans (Charged Off): 13.8%,
loans (Current): 2.8%

avg income for (fully paid): $70,428
avg income for (charged off): $63,516

MTD Loan Applications(forDecember 2021): 4314
MTD Total Funded Amount: $53.98M
MTD Total Amount received: $58.07M

1. Borrowers in categories F and G have a higher average income than the categories (A, B) and considered higher risk and have higher interest rates because their dit ratio is high, indicating a high rate of default.

2. The bank has a general policy of not lending to anyone whose debts exceed 30%, regardless of their income.

3. Category A has the highest percentage of mortgage borrowers, exceeding 51%, which reinforces their classification as the least risky due to their residential stability.

4. Those aged 10+ years of work they are the segment that most frequently requests loans.

5. 1.47% of total loans, representing 18,214 borrowers, are taking out loans to cover previous debts, indicating that the majority of customers are experiencing credit pressures and are looking for a way to manage their payments. The"Credit card repayment" comes in second place with 4,998 borrowers.

6. Over than 60% of loans are "liability settlement" loans. This is a very strong indicator of the bank's risk management.

7. There is a sizable segment that uses loans to invest in fixed assets or improve quality of life, but it remains far less than loans directed towards debt repayment.

8. Educational and environmental loans do not constitute a significant portion of the loan portfolio, which could present an opportunity for the bank to launch marketing campaigns targeting these categories if it wishes to diversify its portfolio. Each represents less than 1%

9. There are 1,776 borrowers for "small business" purposes. This indicates entrepreneurial activity that relies on these loans.

10. Lowest interest rate: starting from 7.35% for category A, and highest interest rate: reaching 21.40% for category G. The difference between each category and the next is approximately 2% to 2.5% consistently and regularly.

11. In grade A, the failure rate is only 5.7%, while it jumps to 31.3% in grade G.

12. Category A is the safety valve for the bank, as it ensures stable and almost guaranteed cash flows.

13. Higher income is statistically linked to a greater ability to repay a loan. Borrowers who fully repay their loans have an average income of around $60,000, while those in default have an average income of around $54,000.

14. Small business loans have the highest default rate, reaching 25.6%.

15. Borrowers for social purposes, demonstrate a higher commitment to repayment compared to those for business or educational purposes, with lower default rates (approximately 9.3% and 9.8%)

16. Loans in renewable energy sector have a high default rate 18.1%, followed by education loans at 15.9% .

17. Car loans and credit cards are performing well with low default rates (around 10.2% – 10.4%)

18. The average income of people who have fully paid off their loans is about $70,428. and The average income of those in default (charged off) is approximately $63,516

19. Borrowers with an annual income approximately 11% higher than others are more likely to successfully complete the repayment process.

20. This figure ($63,516) represents a critical threshold; borrowers whose average income is around or below this amount are statistically inclined to fall into the high-risk category..

21. Total Funded Amount increased significantly throughout the year, which began with $25.3 million in funding in January and ended with $53.98 million in December.

22. The year began with $27.58 million received and 2,332 applications in January and ended with $58.07 million and 4,314 applications in December. This means that the demand for loans increased by nearly 85% in one year.

24. California emerges as the largest loan market by a wide margin, with total financing exceeding $78.4 million, followed by New York with approximately $42 million, and Texas and Florida with approximately $30-31 million each.

25. The bank's activity is primarily concentrated in states with high population density and economic activity.

26. Florida: Despite having the fourth-largest loan portfolio, it suffers from a very high default rate of 17.27%.
Texas : In contrast, Texas has a similar loan portfolio to Florida but a much lower default rate (11.30%).

27. California: Has a moderately high default rate (15.33%). Given that it represents the largest share of the portfolio, any improvement in due diligence procedures there will significantly impact the bank's overall profitability.
New York : Achieves a good balance with a default rate of 12.65%, making it an ideal market for future expansion.

28. Borrowers prefer short-term loans (36 months). This type of loan ensures a faster capital turnover for the bank . Long-term loans (5 years) are larger and targeted at larger projects or needs. For the bank, these loans mean higher "cumulative interest income" over the long term, but they also carry higher risks because predicting a borrower's financial situation after 5 years is more difficult than predicting it after 3 years.

29. The bank lent $273 million (62.7%) as loans ending within 36 months and lent $162.7 million (37.3%) as loans ending within 60 months.

30. The bank places its greatest trust in individuals with long-term job stability (more than 10 years). This job stability reduces the risk of income interruption and thus ensures continued repayment Its total funding amounted to more than $116 million.

31. There is a very high demand for loans from new entrants to the labor market or those at the beginning of their careers, which are the categories that have experience (two years, less than one year, and three years), which come in the next rank with amounts ranging between 44 and 45 million dollars each.
The 6-9 years experience category received the least total funding, with the 9-year category recording the lowest .
amount at approximately $15 million

32. Although the default rate for small projects is the highest (25.6%), the bank lent them $24.1 million.

33. Renters and mortgage holders are the primary drivers of loans at the bank.
The MORTGAGE category received the highest funding with a total of $219.3 million. The RENT category follows closely behind with a total of $185.8 million.
The bank relies almost entirely (over 90% of its funds) on people who do not yet own their homes completely

34. Although a homeowner (OWN) is theoretically considered "safer," the amount of funding available to this category is very low compared to the rest ($29.6 million only).

35. Mortgage holders are not only the most frequent borrowers, but they are also the most likely to request large sums, perhaps due to the bank's confidence in their possession of a "real estate asset" as indirect collateral.

Recomendations

1. To reduce losses, bank must tighten its lending standards for categories D to G, or continue to impose very high interest rates to cover the 31% of loans that are completely lost in those categories.

2. Annual income is not the only factor to guarantee repayment; there are borrowers with high incomes who have also defaulted, which calls for consideration of other factors such as (DTI).

3. Review lending policies for these sectors (Small Business, Educational, House & Moving) or raise interest rates to mitigate the risks

4. The bank can use average income for people who charged off $63,516 as a "guidance threshold" when evaluating new loan applications to reduce default rates

5. The company is expanding rapidly. This accelerated growth in funded amount is excellent news for increasing market share, but it must be ensured that "loan quality" remains high to guarantee that the default rate does not rise in the future in conjunction with this expansion.

Recomendations

6. The bank's ability to collect funds (whether principal or interest payments) is constantly increasing, and this is an excellent indicator of strong cash flow.

7. Marketing expansion: Continued growth indicates that the bank's brand is gaining traction or that there is a growing market demand.

8. Operational pressure: With applications exceeding 4,000 in December, the company must ensure that its credit review team and IT systems can handle this increased volume of applications without delay

9. Texas is considered a "safer and higher-quality" market for banks compared to Florida and other states

10. The company should review its credit policies in Florida and Georgia (where the default rate is above 15%) and conversely, it can increase marketing activity in Texas and Massachusetts because they have low default rates (around 11%)

11. Investing large sums in a high-risk sector(Small Business) significant risk for the bank; which the default rate reached 25.6% and is not small and the probability of losing it is high.

Thank You